

Instructions to the Students:

- Each question carries 12 marks.
- Question No. 1 will be compulsory and include objective-type questions.
- Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
- The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
- Use of non-programmable scientific calculators is allowed.
- Assume suitable data wherever necessary and mention it clearly.

Level/CO Marks

Q. 1 Objective type questions. (Compulsory Question)

12

- The accuracies obtained with the numerical computation of electric fields is usually
(a) 1% (b) 2 to 5% (c) 5 to 10% (d) can be very high
- Most suitable numerical method to solve electrostatic field problems is
(a) Laplace equation method (b) charge simulation method (c) finite difference method (d) resistance analog method
- Streamer mechanism of breakdown explains the phenomena of electrical breakdown of
(a) very short spark gaps (b) when pd is less than 1000 torr-cm (c) very long gaps where field is non-uniform (d) spark gaps subjected to impulse voltages.
- Minimum sparking potential of air is about
(a) 100 V (b) 4.4kV (c) 40 V (d) 325 V
- The breakdown strength of mineral oil is about
(a) 30 kV/mm (b) 50 kV/mm (c) 3 to 5 kV/mm (d) 30 to 40 kV/mm
- Which of the following liquids has the highest breakdown strength?
(a) Mineral oils (b) Silicone oils (c) Chlorinated hydrocarbon oils (d) Polyolefins or esters
- The operating temperatures of polyethylene insulation is
(a) -30° to 50° (b) -60° to 150° (c) -50° to 80° (d) 0° to 100°

8	Electrochemical breakdown and deterioration of insulating material is due to				1
	(a) temperature rise	(b) oxidation, hydrolysis or some other chemical action	(c) only due to hydrolysis and moisture effects	(d) none of the above	
9	For HV cable insulation, the materials used are				1
	(a) glass and ceramic	(b) silicone rubber	(c) XLPE	(d) paper-oil insulation	
10	Gas insulation is nowadays used in				1
	(a) generators	(b) motors	(c) transformers	(d) circuit breakers and substations	
11	The voltage efficiency of a normal impulse generator for generation of switching impulses is				1
	(a) less than 30%	(b) 80 to 90%	(c) 40 to 60%	(d) 10 to 90	
12	Typical capacitive loading on a testing transformer rated for 100 kVA, 250 kV will be about				1
	(a) less than 1 nF	(b) 3 to 5 nF	(c) 10 to 50 nF	(d) 50 to 100 nF	
Q. 2	Solve the following.				12
A)	Explain Poisson's equation in detail.				6
B)	Explain the terms surge voltage and discuss the distribution on it.				6
Q.3	Solve the following.				12
A)	Explain the Streamers theory of breakdown in air at atmospheric pressure.			CO1	6
B)	Explain Townsend's criterion for breakdown in electronegative gases.			CO2	6
Q. 4	Solve Any Two of the following.				12
A)	Discuss the effect of the Following parameters on breakdown strength of liquid			CO1	6
	1) Moisture content in oil				
	2) Solid impurities				
B)	Explain the Different mechanism by which breakdown occurs in solid dielectrics ?			CO2	6
C)	Explain the phenomenon of treeing and tracking.			CO3	6
Q.5	Solve Any Two of the following.				12
A)	Describe impulse voltages standard impulse waves shapes and its equation in detail.			CO1	6

B)	Explain the lightning phenomenon in Detail ?	CO2	6
C)	Write a short note on equipment insulation level & co-ordination of Substations.	CO3	6
Q. 6 Solve Any Two of the following.			12
A)	What are the different non destructive testing are connected to the testing of insulators bushing ,isolators ,C.B. ,Cables, Transformers and surge diverters.	CO1	6
B)	What are the different IES and IS codes adopted for the measurement of high voltage tests on electrical appliances : 1)Power apparatus 2)Electric motor	CO2	6
C)	Write a short note on surge diverter.	CO3	6

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